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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Family Guy
Selected Domain	Domain 2
Selected SDG Track (1–17)	8
Project / Solution Name	Wheat
Name (Leader)	Wu YiWei
Email & Phone (Leader)	era2509029@xmu.edu.my +60 1135449974
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Our project aims to solve problems small business employees might face when manually organizing orders from various sources. They constantly face non-uniform document format, product names and quantity units. These may cause repeated labor, higher rate of faults and difficulty to find the source of problems.

The employees who are directly affected by the problems are those who are incharge of organizing the orders and sales, they have to carry the extra labor and the responsibility of the mistakes. Also, small business owners and decision makers will also be affected since the costs will be higher and work might be delayed.

.On an economic point of view, solving these problems will decrease the cost of labor and increase work efficiency significantly. Also, businesses will be less dependent of personal experience.

As for its societal impacts, after these problems have been solved, the labor and burden of employees will be lessened, the threshold of mastering AI will also be lowered. Small businesses without their own technical support team can also use AI technology in their daily works. Workers can be aware of changes in data immediately, and can use AI to check and correct mistakes in time.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Our proposed application, Wheat, aims to help small business employees organize orders from different sales sources. Employees can upload their order files, a standard product list and a warehouse template, then describe what they need in everyday language. The application will suggest a processing plan for employees to check before running it.

We plan to use a GenAI API to understand different table structures and business instructions. It will suggest how to match column names, identify products and apply

processing rules. For example, it can help employees combine confirmed orders, standardize quantity units and identify possible duplicates. When product names or units are unclear, the application will ask employees to confirm instead of making unsupported decisions.

After confirmation, the program will perform the calculations and generate order details, product summaries and a separate list of exceptions. Processing records will help employees find the original data and understand how each record was handled. The confirmed workflow can be saved and reused the following week. If new files contain unfamiliar statuses or changed fields, the application will highlight them for review.

We also plan to apply skills introduced in the SDC Phase 1 workshops, including GenAI API integration, database design and cloud hosting. The database will store rules, workflow versions and processing records. The prototype is deployed on Replit and can be accessed online through the demonstration link. We also plan to simulate an Orbyte connection for task notifications, employee confirmation and warehouse handover.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Wheat uses React, TypeScript and Vite for its browser interface. Employees upload orders, a product master and an output template, then review the processing plan and results. Python and FastAPI handle the backend. Python CSV parsing and openpyxl read CSV and XLSX files.

Optional AI planning sends bounded file profiles and samples to the Agnes API service using agnes-2.5-flash. The model proposes a constrained JSON workflow. Pydantic and application validation check it before employees approve the business rules. Bundled Python operations perform the transformations, without executing arbitrary model-generated code.

The Replit cloud prototype runs on one Reserved VM with one application worker. Managed PostgreSQL persists browser-scoped workflows, approvals, runs, activity and immutable input/export files. The local desktop entry uses SQLite. SQLAlchemy supports both storage modes. Git and GitHub preserve the source development history.

Replit is an official SDC sponsor resource used for prototype hosting, Status Guard module development, testing and part of the project data analysis. Orbyte task handover is explicitly simulated. Demonstrations use only team-created synthetic business data.

Live prototype: https://wheat-assistant.replit.app

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team plans to use design thinking to understand employees' needs and build a practical solution. We will define the main problem and focus on preparing warehouse documents from different order files. During idea development, we will prepare simple interface sketches . The application will use familiar language, show processing rules clearly and allow employees to confirm or correct them.

For technical feasibility, we will build a prototype and test it with synthetic orders, including missing information and unfamiliar statuses. We will compare the outputs with independently prepared answers.

Before Phase 2, we will improve the deployed prototype, verify cloud access and storage behaviour, test user separation, and prepare instructions for daily use.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Our project mainly supports SDG Target 8.2, which focuses on improving economic productivity through technological upgrading and innovation. Wheat aims to help small business employees reduce repeated work when organizing orders. Source: United Nations, https://sdgs.un.org/goals/goal8.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	For initial planning, we estimate that an employee needs approximately 30 minutes to organize 150 order records from three different sources, including checking mistakes and preparing the warehouse document. This is an assumption rather than a measured result. We will check this estimate through timed manual tests using synthetic orders created by our team.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	We will measure the percentage reduction in employees' active processing time per batch. This includes uploading files, checking rules, correcting unclear records and reviewing results. Initial setup time and system waiting time will be recorded separately. We will also check that the

Metric Category	Your Response
	final results match independently prepared answers, without unexplained missing or repeated records.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Our preliminary target is to reduce employees ' active processing time by at least 30% when reusing a saved workflow. If manual processing takes 30 minutes, our target will be 21 minutes or less. We will test at least three repeat batches with changes in the input files. This is a proposed target, and its feasibility still needs testing.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React, ReactDOM, React type definitions, Vite, @vitejs/plugin-react	Code	MIT
FastAPI, Pydantic, SQLAlchemy, openpyxl Platformdirs	Code	MIT
TypeScript, python-multipart, @playwright/test	Code	Apache-2.0
Uvicorn, HTTPX, psutil, python-dotenv	Code	BSD-3-Clause
Defusedxml	Code	Python Software Foundation License
SQLite	Code	Public domain

pytest, Hatchling	Code	MIT
Hypothesis	Code	MPL-2.0
PyInstaller	Code	GPL-2.0-or-later with bootloader exception
Python	Code	PSF-2.0 and bundled notices
Node.js	Code	MIT and bundled third-party licenses
Uv	Code	MIT OR Apache-2.0
agnes-2.5-flash, accessed through Agnes API	model	Provider API terms

Additional: PostgreSQL (PostgreSQL License), psycopg 3.3.6 (LGPL-3.0-only). PDF tools: PyMuPDF 2.0 (AGPL-3.0/commercial), pypdf 6.10.0 (BSD-3-Clause), Poppler 26.04.0 (GPL-2.0-only OR GPL-3.0-only). Full versions and uses: Technical & Tools Declaration, section 2.2.

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] 

[Date]2026/10/24